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MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF SCIENCE AND TECHNICAL EDUCATION

DEPARTMENT OF MATHEMATICS AND STATISTICS
SEMESTER II, TEST 1, 2023/2024
BACHELOR OF ENGINEERING IN TELECOMMUNICATION SYSTEM
MS 8221: DIFFERENTIAL EQUATION
INSERT UQF 8 (2023/2024)

TIME: 1.00 Hours

Date: 22. 05.2024

SECTION A (15 MARKS)
Answer All questions in this section

QUESTION ONE

Solve the Legendre's equation

$$(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + \mu^2 y = 0.$$

and find the Legendre polynomials $P_n(x)$ for $n = 6$ and $n = 7$.

(5 Marks)

QUESTION TWO

Determine the order of each of the following Bessel's equation

(a)

$$t^2 \frac{d^2 z}{dt^2} + t \frac{dz}{dt} + (\lambda^2 - 64)z = 0.$$

(2.5 Marks)

(b)

$$\sigma^2 \frac{d^2 \eta}{d\sigma^2} + \sigma \frac{d\eta}{d\sigma} + (\Gamma^2 - 625)\eta = 0.$$

(2.5 Marks)

QUESTION THREE

State Rodriguez formula and use it to generate Legendre polynomial $P_n(x)$ for $n = 8$. (5 Marks)

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$$(a), t^2 \frac{d^2 z}{dt^2} + t \frac{dz}{dt} + (\lambda^2 - 64)z = 0,$$

$$\lambda^2 - 64 = 0$$

$$\lambda^2 = 64,$$

$$\lambda = \pm 8.$$

∴ The order is 8,

$$(b), s^2 \frac{d^2 \eta}{ds^2} + s \frac{d\eta}{ds} + (r^2 - 625)\eta = 0.$$

$$r^2 - 625 = 0$$

$$r^2 = 625$$

$$r = \pm 25$$

∴ The order is 25,

Given, $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + 4y = 0 \quad \text{--- (I)}$

$$\frac{d^2 y}{dx^2} - \frac{x}{1-x^2} \frac{dy}{dx} + \frac{4}{1-x^2} y = 0$$

$$P(x) = -\frac{x}{1-x^2} \quad Q(x) = \frac{4}{1-x^2} \quad \Phi(x) = 0$$

at $x=0$,

$P(0) = 0$ $Q(0) = 4$ $\Phi(0) = 0$ hence the point is ordinary point.

$$\text{let } y = \sum_{n=0}^{\infty} a_n x^n \quad \text{--- (i)}$$

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{--- (ii)}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \quad \text{--- (iii)}$$

substitute in eqn (I).

$$1. (1-x^2) \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - x \sum_{n=1}^{\infty} n a_n x^{n-1} + y^2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=2}^{\infty} n(n-1) a_n x^n - \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} y^2 a_n x^n = 0$$

let $n = k+2$ for term one, but $k=n$.

$$\sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k = 2a_2 + 6a_3 x + \sum_{n=2}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

$$2a_2 + 6a_3 x + \sum_{n=2}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=2}^{\infty} n(n-1) a_n x^n - a_1 x - \sum_{n=2}^{\infty} n a_n x^n + y^2 a_0 + y^2 a_1 x +$$

$$\sum_{n=2}^{\infty} y^2 a_n x^n = 0$$

equating L.H.S and R.H.S.

$$2a_2 + y^2 a_0 = 0$$

$$a_2 = -\frac{y^2 a_0}{2}$$

$$6a_3 x - a_1 x + y^2 a_1 x = 0$$

$$6a_3 = a_1 - y^2 a_1$$

$$a_3 = \frac{(1-y^2) a_1}{6}$$

$$(n+2)(n+1) a_{n+2} - n(n-1) a_n - n a_n + y^2 a_n = 0$$

$$(n+2)(n+1) a_{n+2} = (n(n-1) + n - y^2) a_n$$

$$a_{n+2} = \frac{(n(n-1) + n - y^2) a_n}{(n+2)(n+1)}$$

$$n=2 \quad a_4 = \frac{(4-y^2) a_2}{12} = -\frac{y^2(4-y^2) a_0}{24}$$

$$n=3 \quad a_5 = \frac{(9-y^2) a_3}{20} = \frac{(9-y^2)(1-y^2) a_1}{120}$$

$$n=4 \quad a_6 = -\frac{y^2(16-y^2)(4-y^2) a_0}{720}$$

$$n=5 \quad a_7 = \frac{(25-y^2)(9-y^2)(1-y^2) a_1}{5040}$$

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For $y(x) = q_0 + q_1 x + q_2 x^2 + q_3 x^3 + q_4 x^4 + q_5 x^5 + q_6 x^6 + q_7 x^7$

$$y(x) = q_0 + q_1 x - \frac{q_0^2}{2} x^2 + \frac{(1-q_1^2)}{6} q_1 x^3 - \frac{q_1^2(4-q_1^2)}{24} q_0 x^4 + \frac{(7-q_1^2)(1-q_1^2)}{120} q_1 x^5 + \dots$$

$$y(x) = \frac{q_1^2(16-q_1^2)(4-q_1^2)}{720} q_0 x^6 + \frac{(25-q_1^2)(9-q_1^2)(1-q_1^2)}{5040} q_1 x^7$$

$$y(x) = \left[1 - \frac{q_1^2}{2} x^2 - \frac{q_1^2(4-q_1^2)}{24} x^4 - \frac{q_1^2(16-q_1^2)(4-q_1^2)}{720} x^6 \right] q_0 + \left[x + \frac{(1-q_1^2)}{6} x^3 + \frac{(7-q_1^2)(1-q_1^2)}{120} x^5 + \frac{(25-q_1^2)(9-q_1^2)(1-q_1^2)}{5040} x^7 \right] q_1$$

Then $P_n(x)$

for $n=6$:

$$y(x) = \left[1 - \frac{q_1^2}{2} x^2 - \frac{q_1^2(4-q_1^2)}{24} x^4 - \frac{q_1^2(16-q_1^2)(4-q_1^2)}{720} x^6 \right] q_0$$

$$P_6(x) = q_0 \left[1 - \frac{q_1^2}{2} x^2 - \frac{q_1^2(4-q_1^2)}{24} x^4 - \frac{q_1^2(16-q_1^2)(4-q_1^2)}{720} x^6 \right]$$

for $n=7$:

$$P_7(x) = q_1 \left[x + \frac{(1-q_1^2)}{6} x^3 + \frac{(7-q_1^2)(1-q_1^2)}{120} x^5 + \frac{(25-q_1^2)(9-q_1^2)(1-q_1^2)}{5040} x^7 \right]$$



3 Rodriguez formula state that

$$P_n(x) = \frac{1}{2^n \cdot n!} \frac{d^n}{dx^n} (x^2-1)^n$$

for $n=8$,

$$P_8(x) = \frac{1}{2^8 \cdot 8!} \frac{d^8}{dx^8} (x^2-1)^8$$

$$P_8(x) = \left(\frac{1}{2^8 \cdot 8!} \right) \frac{d^7}{dx^7} 16x(x^2-1)^7$$

$$P_8(x) = \left(\frac{16}{2^8 \cdot 8!} \right) \frac{d^6}{dx^6} [(x^2-1)^7 + 14x^2(x^2-1)^6]$$

$$P_8(x) = \left(\frac{16}{2^8 \cdot 8!} \right) \frac{d^5}{dx^5} [14x(x^2-1)^6 + 14(2x(x^2-1)^6 + 12x^3(x^2-1)^5)]$$

$$P_8(x) = \left(\frac{16 \times 14}{2^8 \cdot 8!} \right) \frac{d^4}{dx^4} [(x^2-1)^6 + 12x(x^2-1)^5 + (2(x^2-1)^6 + 24x(x^2-1)^5 + 36x^2(x^2-1)^5 + 120x^4(x^2-1)^4)]$$

$$P_8(x) =$$